

# Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics<sup>1</sup>

## Supplemental Material

### A Proof of Proposition 4 (aggregate uncertainty and the conditional steady states)

Let  $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$  denote the gross return,  $c(K, A)$  the stochastic-equilibrium consumption function, and  $c^{nul}(A, K) := AK^\alpha - \delta K$  the conditional  $K$ -nullcline (shorthand for the main text's  $C_A^{Kcnull}(K)$ ). As in the conditional steady-state Euler equation of Section 2.2, define the conditional Euler residuals at candidate  $K$ -stationary points, using equilibrium post-switch consumption  $c(K, A')$ :

$$\begin{aligned} F_B(K) &:= \beta \left[ \pi_{BB} R(B, K) + \pi_{BG} \frac{c^{nul}(B, K)}{c(K, G)} R(G, K) \right] - 1, \\ F_G(K) &:= \beta \left[ \pi_{GG} R(G, K) + \pi_{GB} \frac{c^{nul}(G, K)}{c(K, B)} R(B, K) \right] - 1. \end{aligned} \quad (1)$$

By that Euler equation,  $F_B(K_B^{cs}) = 0$  and  $F_G(K_G^{cs}) = 0$ .

**Proposition A.1** (Aggregate uncertainty and the conditional steady states; restatement of Proposition 4).

*Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),*

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})}; \quad (2)$$

*and (ii) each conditional Euler residual  $F_A$  in (1) is strictly positive for  $K < K_A^{cs}$  and strictly negative for  $K > K_A^{cs}$  on the relevant range (single crossing). Then*

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}. \quad (3)$$

*Proof.*

At any conditional or PF steady state, consumption lies on the nullcline:  $c_A^{cs} = c(K_A^{cs}, A) = c^{nul}(A, K_A^{cs})$  and  $c_A^{pf} = c^{nul}(A, K_A^{pf})$ .

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The PF steady states solve the Euler equations under absorbing beliefs,  $1 = \beta R(B, K_B^{pf})$  and  $1 = \beta R(G, K_G^{pf})$ . Since  $R(A, K)$  is strictly increasing in  $A$  and strictly decreasing in  $K$  (as  $\alpha - 1 < 0$ ), immediately  $K_B^{cs} < K_G^{pf}$ ; this step is belief-free and uses neither hypothesis. Hypothesis (ii) makes  $K_B^{cs}, K_G^{cs}$  the unique zero crossings of  $F_B, F_G$ , from above.

*Step 1: show  $K_B^{cs} < K_B^{pf}$ .* Evaluate  $F_B$  at  $K_B^{pf}$ , using  $1 = \beta R(B, K_B^{pf})$ ,  $\pi_{BB} = 1 - \pi_{BG}$ , and  $c^{nul}(B, K_B^{pf}) = c_B^{pf}$ :

$$\begin{aligned} F_B(K_B^{pf}) &= \beta \left[ (1 - \pi_{BG}) R(B, K_B^{pf}) + \pi_{BG} \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) \right] - 1 \\ &= \pi_{BG} \left[ \beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) - 1 \right]. \end{aligned} \quad (4)$$

Jump dominance (2) gives  $c(K_B^{pf}, G) > c_B^{pf} R(G, K_B^{pf}) / R(B, K_B^{pf})$ , hence

$$\beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) < \beta R(B, K_B^{pf}) = 1,$$

so the bracket in (4) is strictly negative and  $F_B(K_B^{pf}) < 0$ . As  $F_B$  is negative only above its zero,  $K_B^{cs} < K_B^{pf}$ .

*Step 2: show  $K_G^{cs} > K_G^{pf}$ .* Symmetrically, using  $1 = \beta R(G, K_G^{pf})$ ,  $\pi_{GG} = 1 - \pi_{GB}$ , and  $c^{nul}(G, K_G^{pf}) = c_G^{pf}$ ,

$$F_G(K_G^{pf}) = \pi_{GB} \left[ \beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) - 1 \right].$$

Jump dominance (2) gives  $c(K_G^{pf}, B) < c_G^{pf} R(B, K_G^{pf}) / R(G, K_G^{pf})$ , hence

$$\beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) > \beta R(G, K_G^{pf}) = 1,$$

so  $F_G(K_G^{pf}) > 0$ . As  $F_G$  is positive only below its zero,  $K_G^{pf} < K_G^{cs}$ .

*Step 3: conclude the ordering and translate to consumption.* Steps 1 and 2 give  $K_B^{cs} < K_B^{pf}$  and  $K_G^{pf} < K_G^{cs}$ , and already  $K_B^{pf} < K_G^{pf}$ , hence  $K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}$ . The  $K$ -nullcline  $c^{nul}(A, K) = AK^\alpha - \delta K$  is strictly increasing in  $A$  and increasing in  $K$  on the dynamically efficient region ( $\alpha AK^{\alpha-1} > \delta$ , which holds at all four steady states under

standard calibrations), so the same ordering carries over:

$$c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}.$$

■

**Reading the hypotheses** Jump dominance (2) compares two responses to the same regime switch at a PF steady state. The return ratio deviates from one only at first order in the regime gap and with a small coefficient,  $R(G, K)/R(B, K) = 1 + (G - B) \alpha K^{\alpha-1} / R(B, K)$ , whereas equilibrium consumption responds to a persistent switch at first order with an  $O(1)$  coefficient. Both hypotheses are checkable from the computed decision rules: (i) by evaluating  $c(\cdot, A')$  at the two PF steady states, (ii) by evaluating  $F_A$  on a grid of  $K$ . In the baseline quarterly calibration ( $\beta = 0.99$ ,  $\alpha = 0.33$ ,  $\delta = 0.025$ ,  $A \in \{0.99, 1.01\}$ ,  $\pi_{AA} = 0.90$ ) both hold with wide margins: at  $K_B^{pf}$  consumption rises by 0.77% upon a switch to  $G$  against a return-ratio threshold of 0.07%, and at  $K_G^{pf}$  it falls by 0.76%. Each residual  $F_A$  crosses zero exactly once on the computed saddle range, from above. The implied orderings are confirmed in the computed equilibrium,  $K_B^{cs} = 27.53 < K_B^{pf} = 27.93 < K_G^{pf} = 28.77 < K_G^{cs} = 29.22$  (the anticipated-reversion figure, main text Section 2.2).

**Necessity of the strict conditions** The hypotheses cannot be dropped. Under full depreciation ( $\delta = 1$ ) with log utility and Cobb–Douglas production, the equilibrium policy takes the Brock and Mirman (1972) form  $K' = \alpha \beta A K^\alpha$  with  $c = (1 - \alpha \beta) A K^\alpha$ , independently of  $\Pi$ . Upon a switch at fixed  $K$ , consumption then moves exactly by  $G/B$ , and so does  $R(A, K) = \alpha A K^{\alpha-1}$ : jump dominance holds with equality, and  $K_A^{cs} = K_A^{pf} = (\alpha \beta A)^{1/(1-\alpha)}$ . The strict orderings therefore require strict jump dominance; away from this degenerate corner the conditions hold in the calibrated equilibrium and the strict inequalities obtain.

## B Proof of Proposition 6 (state-dependent peak timing)

*Proof.*

We derive the peak time in closed form and differentiate it to sign the comparative static in initial capital.

*Step 1 (Blanchard–Kahn solution on the saddle branch).* Log-linearize the RBC dynamics around the non-stochastic steady state, with  $\hat{k}_t := \ln K_t - \ln K^*$  and  $\hat{a}_t := \ln A_t - \ln A^*$ . The

linearized capital law of motion and saddle-path consumption policy are

$$\hat{k}_{t+1} = \lambda \hat{k}_t + \theta \hat{a}_t, \quad (5)$$

$$\hat{c}_t = \psi_k \hat{k}_t + \psi_a \hat{a}_t, \quad (6)$$

where  $\lambda \in (0, 1)$  is the stable eigenvalue,  $\theta > 0$  the capital-TFP elasticity, and  $\psi_k > 0$ ,  $\psi_a > 0$  the consumption elasticities on the saddle branch. For a one-time shock  $\hat{a}_0 = \varepsilon > 0$  with  $\hat{a}_t = \rho^t \varepsilon$ ,  $\rho \in (0, 1)$ , iterating (5) gives (using  $\lambda \neq \rho$ )

$$\hat{k}_t = \lambda^t \hat{k}_0 + \theta \varepsilon \sum_{s=0}^{t-1} \lambda^{t-1-s} \rho^s = \lambda^t \hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} (\lambda^t - \rho^t). \quad (7)$$

Substituting (7) into (6) yields  $\hat{c}_t = \mathcal{P} \lambda^t + \mathcal{Q} \rho^t$ , where

$$\mathcal{P}(\hat{k}_0) := \psi_k \left[ \hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} \right], \quad \mathcal{Q} := \left[ \psi_a - \frac{\psi_k \theta}{\lambda - \rho} \right] \varepsilon. \quad (8)$$

Crucially,  $\mathcal{P}$  depends on  $\hat{k}_0$  while  $\mathcal{Q}$  does not.

*Step 2 (Peak time in closed form).* Treating  $t$  as continuous, the peak satisfies  $d\hat{c}_t/dt = 0$ , i.e.  $\mathcal{P} \ln \lambda \cdot \lambda^{t^*} + \mathcal{Q} \ln \rho \cdot \rho^{t^*} = 0$ , so  $(\lambda/\rho)^{t^*} = -\mathcal{Q} \ln \rho / (\mathcal{P} \ln \lambda)$ . Taking logarithms,

$$t^* = \frac{1}{\ln(\lambda/\rho)} \ln \left( \frac{-\mathcal{Q} \ln \rho}{\mathcal{P} \ln \lambda} \right), \quad (9)$$

well-defined when  $-\mathcal{Q} \ln \rho / (\mathcal{P} \ln \lambda) > 0$ . Since  $\ln \lambda < 0$  and  $\ln \rho < 0$  cancel, positivity requires  $\mathcal{Q}/\mathcal{P} < 0$ ; with  $\mathcal{P} > 0$  this means  $\mathcal{Q} < 0$ , i.e.  $\psi_a < \psi_k \theta / (\lambda - \rho)$ : the direct TFP effect is dominated by the capital-accumulation channel, which generates the hump.

*Step 3 (Comparative static).* Differentiating (9) directly, and using  $\partial \mathcal{P} / \partial \hat{k}_0 = \psi_k > 0$  from (8),

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{1}{\ln(\lambda/\rho)} \cdot \frac{-1}{\mathcal{P}} \cdot \frac{\partial \mathcal{P}}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)}. \quad (10)$$

Under  $\lambda > \rho$  we have  $\ln(\lambda/\rho) > 0$ ; combined with  $\psi_k > 0$  and  $\mathcal{P} > 0$  (which holds when  $\hat{k}_0 > -\theta \varepsilon / (\lambda - \rho)$ , easily satisfied within the ergodic capital range), this gives  $\partial t^* / \partial \hat{k}_0 < 0$ . That is, the consumption peak arrives strictly later when initial capital is farther below the steady state. ■

## C Proofs of Theorem 2 and Proposition 9 (synchronization and the across-spell bound)

### Proof of Theorem 2 (synchronization under frozen-regime contraction)

*Proof.*

*Part (i): existence of the long-history limit.* Fix a regime history  $\omega = (\dots, A_{t-2}, A_{t-1})$  and, for  $T \geq 0$ , let  $G_T := \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T-1}} : \mathcal{U} \rightarrow \mathcal{U}$  (forward invariance). For  $T' > T$  and  $\Phi, \Phi' \in \mathcal{U}$ , writing  $\Psi := (\Gamma_{A_{t-T-2}} \circ \dots \circ \Gamma_{A_{t-T'-1}})(\Phi') \in \mathcal{U}$ ,

$$d(G_T(\Phi), G_{T'}(\Phi')) = d(G_T(\Phi), G_T(\Psi)) \leq \lambda_c^{T+1} \text{diam}(\mathcal{U}), \quad (11)$$

since  $G_T$  composes  $T + 1$  maps each contracting at rate  $\lambda_c$ . Thus  $\{G_T(\Phi)\}$  is Cauchy uniformly in  $\Phi$ , and  $\Phi^*(\omega) := \lim_{T \rightarrow \infty} G_T(\Phi)$  exists, is independent of  $\Phi$ , and lies in  $\mathcal{U}$  ( $\mathcal{U}$  closed in the complete  $(\mathcal{X}, d)$ ). Uniqueness: any  $\tilde{\Phi} \in \mathcal{U}$  consistent with  $\omega$  (i.e.  $\tilde{\Phi} = G_T(\tilde{\Phi}_{t-T-1})$  for all  $T$ ) satisfies  $d(\tilde{\Phi}, \Phi^*(\omega)) \leq \lambda_c^{T+1} \text{diam}(\mathcal{U})$  for every  $T$ , hence  $\tilde{\Phi} = \Phi^*(\omega)$ ; this is the random-fixed-point construction of [Schenk-Hoppé and Schmalfuß \(2001\)](#) for finitely many contracting maps selected by a Markov chain. With  $\Gamma^{(T)} := \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T}}$ , the identity  $\Phi^*(\omega) = \Gamma^{(T)}(\Phi^*(\sigma^{-T}\omega))$  and rate- $\lambda_c^T$  contraction give

$$d(\Gamma^{(T)}(\Phi), \Phi^*(\omega)) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (12)$$

*Part (ii): suffix synchronization.* If  $\omega, \omega'$  agree over the most recent  $T$  periods with common composition  $\Gamma^{(T)}$ , then  $\Phi^*(\omega) = \Gamma^{(T)}(\Phi^*(\sigma^{-T}\omega))$  (and likewise  $\omega'$ ), so with  $\Phi^*(\sigma^{-T}\omega), \Phi^*(\sigma^{-T}\omega') \in \mathcal{U}$ ,

$$d(\Phi^*(\omega), \Phi^*(\omega')) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (13)$$

■

### Proof of Proposition 9 (across-spell approximation bound)

*Proof.*

*Step 1 (Pathwise bound).* By Theorem 2(i),  $\Phi = \Phi^*(\omega)$ ,  $\Phi' = \Phi^*(\omega')$ ; if the histories agree over the most recent  $T$  periods, Theorem 2(ii) gives  $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$ . This

pathwise bound survives conditioning on any subset of realizations, in particular fixing a common  $K$ , so it holds for every pair in the same- $K$  cloud.

*Step 2 (Distribution of the common suffix length).* Draw the two histories independently from the stationary regime chain, conditioned to agree at the current date. That shared regime indexes the cloud but has not yet acted on  $\Phi$ , so the count of shared maps begins one period earlier. The symmetric two-state chain has uniform stationary distribution, hence is reversible with the same reversed transition matrix. Stepping backward, the two reversed chains still agree if both persist ( $\rho_A^2$ ) or both switch ( $((1 - \rho_A)^2)$ ), independently across periods, so with

$$q := \rho_A^2 + (1 - \rho_A)^2 \in (1/2, 1), \quad (14)$$

$$\Pr(T = k) = (1 - q)q^k, \quad k = 0, 1, 2, \dots$$

*Step 3 (Expected cloud diameter).* Combining Steps 1–2,

$$\mathbb{E}[d(\Phi, \Phi')] \leq \text{diam}(\mathcal{U}) \sum_{k=0}^{\infty} (1 - q)q^k \lambda_c^k = \frac{1 - q}{1 - q\lambda_c} \text{diam}(\mathcal{U}), \quad (15)$$

convergent since  $q\lambda_c < 1$ . The factor  $\frac{1-q}{1-q\lambda_c}$  in (15) is below one and vanishes as  $\rho_A \rightarrow 1$  ( $q \rightarrow 1$ ). It does not vanish as  $\lambda_c \rightarrow 0$ , because a pair sharing no regime history is bounded only by  $\text{diam}(\mathcal{U})$ , an event of probability  $1 - q$ . ■

## D Proof of Proposition 10 (local contraction rate on the conditional saddle)

**Regularity conditions** Fix  $A \in \{B, G\}$  and assume at the conditional steady state  $\Phi_A^{cs}$ :

- (i) Near  $\Phi_A^{cs}$ ,  $\mathcal{X}$  is an open subset of a Banach space with norm  $\|\cdot\|$  and  $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$  is  $C^1$ ; write  $D\Gamma_A := D\Gamma_{\text{endo}}(\Phi_A^{cs}, A)$ .
- (ii) The tangent to the conditional saddle  $\mathcal{M}(\Phi_0, A)$  at  $\Phi_A^{cs}$  is a one-dimensional invariant direction of  $D\Gamma_A$ : the span of an eigenvector with simple real eigenvalue  $\lambda_A^s \in (0, 1)$ , separated from the rest of the spectrum by a spectral gap.
- (iii) In a neighborhood of  $\Phi_A^{cs}$  there is a  $C^1$  one-dimensional locally invariant curve  $\mathcal{W}_A \subset \mathcal{X}$ , tangent at  $\Phi_A^{cs}$  to that direction, with  $\mathcal{M}(\Phi_0, A) \cap \mathcal{V} \subset \mathcal{W}_A$  for some neighborhood  $\mathcal{V}$  of  $\Phi_A^{cs}$ .

- (iv) The metric  $d$  is asymptotically equivalent to  $\|\cdot\|$  near  $\Phi_A^{cs}$ :  $d(\Phi, \Phi')/\|\Phi - \Phi'\| \rightarrow 1$  as  $\Phi, \Phi' \rightarrow \Phi_A^{cs}$ ,  $\Phi \neq \Phi'$ .
- (v) The aggregate capital functional  $K : \mathcal{X} \rightarrow \mathbb{R}$  is  $C^1$  near  $\Phi_A^{cs}$  and nondegenerate along the stable direction,  $DK[e_A^s] \neq 0$ .

Conditions (i)–(iv) deliver the contraction rate; (v) is used only for the capital recursion (17), on which every conclusion stated in terms of  $\{K_t\}$  rests.

**Full statement** Under (i)–(iv), for  $\Phi, \Phi'$  restricted to  $\mathcal{M}(\Phi_0, A)$ ,

$$\lim_{\substack{\Phi, \Phi' \rightarrow \Phi_A^{cs} \\ \Phi, \Phi' \in \mathcal{M}(\Phi_0, A), \Phi \neq \Phi'}} \frac{d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A))}{d(\Phi, \Phi')} = |\lambda_A^s|, \quad (16)$$

and for every  $\varepsilon > 0$  there is a neighborhood  $\mathcal{U}_A^\varepsilon$  of  $\Phi_A^{cs}$  on which  $\Gamma_{\text{endo}}(\cdot, A)$  contracts  $\mathcal{M}(\Phi_0, A)$  at rate  $|\lambda_A^s| + \varepsilon$ .

**Aggregate-capital projection** Under (v),  $\mathcal{W}_A$  and the orbit tail it contains are locally parametrized by  $K$  near  $K_A^{cs}$ , and

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|), \quad (17)$$

exact as  $K_t \rightarrow K_A^{cs}$ , the form quoted in main-text Proposition 10. The sign in (17) is not delivered by (16), which fixes only the magnitude. It comes from the proof's expansion at  $s' = 0$ :  $\Gamma_A \phi(s) - \Phi_A^{cs} = \lambda_A^s s e_A^s + o(s)$ , while local invariance writes  $\Gamma_A \phi(s) = \phi(\tilde{s})$ , so  $\tilde{s} = \lambda_A^s s + o(s)$ . With  $\kappa := DK[e_A^s] \neq 0$  from (v),  $K_t - K_A^{cs} = \kappa s_t + o(s_t)$ , and (17) follows with its sign since  $\lambda_A^s > 0$ .

*Proof.* Work in the Banach identification of (i) and write  $\Gamma_A := \Gamma_{\text{endo}}(\cdot, A)$ ; by (ii) let  $e_A^s$  be a unit eigenvector with  $D\Gamma_A e_A^s = \lambda_A^s e_A^s$ . By (iii) take a  $C^1$  parametrization  $\phi : (-\eta, \eta) \rightarrow \mathcal{W}_A$  with  $\phi(0) = \Phi_A^{cs}$ ,  $\phi'(0) = e_A^s$  carrying the orbit tail; for  $\Phi = \phi(s)$ ,  $\Phi' = \phi(s')$ ,  $s \neq s'$ ,

$$\Phi - \Phi' = (s - s') e_A^s + r_1(s, s'), \quad \|r_1(s, s')\| = o(|s - s'|). \quad (18)$$

By (i), continuity of  $D\Gamma_{\text{endo}}$  near  $\Phi_A^{cs}$  gives the uniform two-point expansion  $\Gamma_A \Phi - \Gamma_A \Phi' = D\Gamma_A(\Phi - \Phi') + r_2$  with  $\|r_2\| = o(\|\Phi - \Phi'\|)$ . Substituting (18) and  $D\Gamma_A e_A^s = \lambda_A^s e_A^s$  yields

$\Gamma_A \Phi - \Gamma_A \Phi' = \lambda_A^s (s - s') e_A^s + o(|s - s'|)$ ; taking norms with  $\|\Phi - \Phi'\| = |s - s'| + o(|s - s'|)$  from (18),

$$\frac{\|\Gamma_A \Phi - \Gamma_A \Phi'\|}{\|\Phi - \Phi'\|} \longrightarrow |\lambda_A^s| \quad \text{as } s, s' \rightarrow 0, s \neq s'. \quad (19)$$

By (iv),  $d(\Phi, \Phi')/\|\Phi - \Phi'\| \rightarrow 1$  applied to numerator and denominator of (19) gives (16). For the neighborhood statement, fix  $\varepsilon > 0$ ; the limit (16) yields  $\delta_\varepsilon > 0$  such that  $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$  with  $d(\Phi, \Phi_A^{cs}), d(\Phi', \Phi_A^{cs}) < \delta_\varepsilon$  imply  $d(\Gamma_A \Phi, \Gamma_A \Phi') \leq (|\lambda_A^s| + \varepsilon) d(\Phi, \Phi')$ ; take  $\mathcal{U}_A^\varepsilon := B_{\delta_\varepsilon}(\Phi_A^{cs})$ . ■

## E Primitive conditions for Assumption 2

This appendix gives an independent route to Assumption 2 from static primitives at  $\mu = 1$  (rational expectations), without the homotopy of Appendix F: side-switching across the  $K$ -nullcline is excluded under MPC bounds and weak price feedback. Near the conditional steady state local spectral regularity alone gives exact  $K$ -sufficiency (part (a)). Away from it the log–Cobb–Douglas structure reduces the verification chain to explicit interval inequalities organized by a  $K$ -monotonicity interval and a  $K$ -admissible continuation, checked on a computed continuation rather than from primitives (part (b)).

Two objects organize the hypotheses. An interval  $[\underline{K}, \overline{K}]$  with  $0 < \underline{K}$  and  $\overline{K} \leq (A/\delta)^{1/(1-\alpha)}$  is a  $K$ -monotonicity interval with margin  $\kappa \in (0, 1)$  if  $B_A(\underline{K}) - N_A(\overline{K}) \leq 1 - \kappa$  and  $2f(\overline{K}; A) + (1 - \delta)\overline{K} < 8\kappa/M_A(\underline{K})$ , where  $B_A, N_A, M_A$  are the MPC–price-feedback, nullcline-slope, and nullcline-curvature functions given in the proof. A frozen- $A$  continuation is  $K$ -admissible on  $(\mathcal{K}, \mathcal{V}_A)$  if (i) its  $K$ -path and the segments joining consecutive  $(K_t, K_{t+1})$  stay in  $\mathcal{K}$ , and (ii) every  $K$ -stationary state with capital in  $\mathcal{K}$  lies in  $\mathcal{V}_A$ .

**Theorem E.1** (Within-spell  $K$ -sufficiency under log utility and Cobb–Douglas).

Fix  $A \in \{B, G\}$  and a conditional saddle  $\mathcal{M}(\Phi_0, A)$  in the heterogeneous-household economy of this section.

- (a) Suppose the regularity conditions of Proposition 10 hold at  $\Phi_A^{cs}$  with stable eigenvalue  $\lambda_A^s \in (0, 1)$ . Then there exists a neighborhood  $\mathcal{V}_A$  of  $\Phi_A^{cs}$  such that, along the frozen- $A$  continuation started from any  $\Phi'_0 \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$ : (i)  $\{K_t\}$  is strictly monotone and converges to  $K_A^{cs}$  without crossing it; (ii)  $\Delta K_t = 0$  only at  $\Phi_t = \Phi_A^{cs}$ ; and (iii)  $K$  is an exact conditional sufficient statistic on  $\mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$ .



(b) Suppose, in addition, that Assumptions 3 and 4 hold, and let  $\mathcal{K}$  be a  $K$ -monotonicity interval with margin  $\kappa \in (0, 1)$ . Then along every continuation that is  $K$ -admissible on  $(\mathcal{K}, \mathcal{V}_A)$ , with  $\mathcal{V}_A$  as in part (a), Assumption 2 holds,  $\{K_t\}$  is strictly monotone until convergence, and  $K$  is an exact conditional sufficient statistic on the entire continuation.

Part (a) (proximity measured in the distributional state space) covers the late phase of a spell, part (b) the full ergodic range; under the [Krusell and Smith \(1998\)](#) calibration the ergodic range is a  $K$ -monotonicity interval with margin  $\kappa = 0.95$  (verified below). Part (b) is not assumption-free: Assumption 4's derivation from Assumption 3 is open, and admissibility (ii) (no plateau outside the local neighborhood) is the single residual condition separating it from an unconditional global theorem.

**Household MPC bounds** Let  $c(a, z; r, w) = \tilde{c}(x, z)$  be the stationary consumption policy, entering prices only through cash-on-hand  $x = wz + (1 + r)a$ . Along a frozen-regime continuation the relevant bound is Lipschitz in price-induced cash-on-hand changes, not a transfer MPC.

**Assumption 3** (Bounded price-income MPC).

Along the frozen-regime continuation under  $A$ , there exists  $\bar{m} \in (0, 1]$  such that, for all  $(a, z)$  in the support of  $\Phi_t$ , all  $t$ , and all price-induced cash-on-hand levels  $x' \geq x$  along the continuation,<sup>2</sup>

$$0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq \bar{m}(x' - x). \quad (20)$$

Normality of consumption and saving in  $x$  gives  $0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq x' - x$ , so Assumption 3 holds with  $\bar{m}_{\text{stat}} = 1$  and needs no calibration or smoothness. The pathwise constant  $\bar{m}_{\text{path}}$  of Assumption 4 is a distinct object, capped at one by that assumption; the pointwise bound below is stated in it.

Define  $\bar{Z} := \sup_{t \geq 0} \int z d\Phi_t$ ,  $\bar{a} := \sup_{t \geq 0} \int a d\Phi_t$ , and the *segment hull*  $\mathcal{K}^{\text{seg}} := \bigcup_{t \geq 0} [\min\{K_t, K_{t+1}\}, \max\{K_t, K_{t+1}\}]$ .

**Two-step verification of Assumption 2** Lemma [E.1](#) gives a sufficient condition for *endpoint no-side-switching*,  $D_t D_{t+1} > 0$  for every transition with  $K_{t+1} \neq K_t$ ; Lemma [E.2](#)

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<sup>2</sup>Bounds *price-induced* income changes, not the transfer MPC; the Lipschitz form accommodates kinks from discrete idiosyncratic states, and precautionary saving pushes  $\bar{m}$  below one ([Kimball, 1990](#)). [Kaplan and Violante \(2014\)](#) estimate quarterly MPCs of 0.15–0.25, though nothing here depends on these values.

upgrades this to the full line-segment no-crossing of Assumption 2 via a curvature-margin check. Both are numerically checkable.

In the Krusell–Smith Cobb–Douglas benchmark, with  $A$  the effective productivity of Section 4.1 and  $L_A$  the regime’s aggregate labor,  $w_A(K) = (1 - \alpha)AK^\alpha/L_A$  and  $r_A(K) = \alpha AK^{\alpha-1} - \delta$ , so  $w'_A(K) = (1 - \alpha)\alpha AK^{\alpha-1}/L_A$ ,  $r'_A(K) = \alpha(\alpha - 1)AK^{\alpha-2}$ ,  $(C_A^{Kcnull})'(K) = \alpha AK^{\alpha-1} - \delta$ . The first step needs a path-level finite-difference bound.

**Assumption 4** (Pathwise finite-difference bound on aggregate consumption).

*Along the frozen- $A$  continuation, there exists  $\bar{m} \in (0, 1]$  such that for every  $t$  with  $K_{t+1} \neq K_t$ ,*

$$|C_{t+1} - C_t| \leq \bar{m}(w'_A(\xi_t)\bar{Z} + |r'_A(\xi_t)|\bar{a})|K_{t+1} - K_t| \quad (21)$$

for some  $\xi_t$  between  $K_t$  and  $K_{t+1}$ .

Assumption 4 is the discrete-time aggregate analogue of Assumption 3 but logically distinct: it additionally requires distributional-composition and continuation-value effects not to exceed the static bound. The implied  $\bar{m}_{\text{path}} \approx 0.03$  on the simulated Krusell and Smith (1998) path sits below the  $\bar{m}_{\text{path}} = 1$  envelope of Proposition E.1 (gap reflecting the small constrained mass); deriving it from Assumption 3 is left to future work.

**Lemma E.1** (Primitive condition for endpoint no-side-switching).

*Fix  $A \in \{B, G\}$  and the frozen- $A$  continuation with  $K_t := K(\Phi_t)$ ,  $C_t := C(\Phi_t, A)$ . Suppose Assumption 4 holds and*

$$\sup_{K \in \mathcal{K}^{\text{seg}}} \bar{m}(w'_A(K)\bar{Z} + |r'_A(K)|\bar{a}) - \inf_{K \in \mathcal{K}^{\text{seg}}} (C_A^{Kcnull})'(K) < 1. \quad (22)$$

*Then endpoint no-side-switching holds:  $D_t D_{t+1} > 0$  for every transition with  $K_{t+1} \neq K_t$ . Strict monotonicity additionally requires unique  $K$ -stationarity (main text), ruling out nonterminal flat steps with  $K_{t+1} = K_t$  at  $\Phi_t \neq \Phi^{cs}$ .*

*Proof.*

With  $D_t := C_t - C_A^{Kcnull}(K_t)$  we have  $K_{t+1} - K_t = -D_t$ . The mean-value theorem gives  $C_A^{Kcnull}(K_{t+1}) - C_A^{Kcnull}(K_t) = N_t(K_{t+1} - K_t)$ ,  $N_t := (C_A^{Kcnull})'(\xi_t^N)$ , and (21) gives  $|C_{t+1} - C_t| \leq B_t|D_t|$ ,  $B_t := \bar{m}(w'_A(\xi_t)\bar{Z} + |r'_A(\xi_t)|\bar{a})$ . Then  $D_{t+1} = D_t(1 + N_t) + (C_{t+1} - C_t)$ , so  $|D_{t+1} - D_t(1 + N_t)| \leq B_t|D_t|$ . Since  $\xi_t, \xi_t^N \in \mathcal{K}^{\text{seg}}$ , (22) gives  $B_t - N_t < 1$ , i.e.  $1 + N_t - B_t > 0$ ; hence  $D_{t+1} \geq D_t(1 + N_t - B_t) > 0$  for  $D_t > 0$  and  $D_{t+1} \leq D_t(1 + N_t - B_t) < 0$  for  $D_t < 0$ , so  $D_t D_{t+1} > 0$  whenever  $K_{t+1} \neq K_t$ . Strict monotonicity further needs  $K_{t+1} = K_t$  only

at  $\Phi_t = \Phi^{cs}(\Phi_0, A)$ , by Assumption 1(i) (representative-agent) or the stable eigenvalue  $\lambda_A^s \in (0, 1)$  (heterogeneous-agent). ■

Consecutive endpoints stay on the same side, but the straight segment can still cross the curved nullcline; the next lemma closes this gap.

**Lemma E.2** (From endpoint no-side-switching to Assumption 2).

Let  $N : K \mapsto N(K)$  be  $C^2$  on  $[K_t, K_{t+1}]$  with  $|N''(K)| \leq M$ , and set  $D_\tau := C_\tau - N(K_\tau)$  for  $\tau \in \{t, t+1\}$ . If  $D_t D_{t+1} > 0$  and

$$\min\{|D_t|, |D_{t+1}|\} > \frac{M(K_{t+1} - K_t)^2}{8}, \quad (23)$$

then the straight line segment from  $(K_t, C_t)$  to  $(K_{t+1}, C_{t+1})$  does not intersect the graph of  $N$ .

*Proof.*

With  $L$  the segment and  $G := L - N$ ,  $G(K_t) = D_t$ ,  $G(K_{t+1}) = D_{t+1}$ ,  $|G''| \leq M$ . Let  $L_D$  linearly interpolate  $(K_t, D_t)$ ,  $(K_{t+1}, D_{t+1})$ ; the quadratic interpolation-error bound gives  $\sup_{[K_t, K_{t+1}]} |G - L_D| \leq M(K_{t+1} - K_t)^2/8$ . Since  $D_t, D_{t+1}$  share a sign,  $L_D$  is monotone with  $|L_D| \geq \min\{|D_t|, |D_{t+1}|\}$ , so (23) gives  $|G - L_D| < |L_D|$  pointwise; thus  $G$  keeps that sign and  $L \neq N$  on  $[K_t, K_{t+1}]$ . ■

Lemmas E.1 and E.2 together verify Assumption 2, both numerically checkable. In the baseline Krusell and Smith (1998) calibration  $|D_t| = |K_{t+1} - K_t| \sim 10^{-2}$  and  $M \sim 10^{-3}$ , so  $M(K_{t+1} - K_t)^2/8 \approx 10^{-8} \ll |D_t|$ .

**Hierarchy of primitive sufficient conditions** Proposition E.1 reduces (22) to a substitution at the conditional steady state under log utility and Cobb–Douglas. With the order-preservation diagnostic (Proposition G.1) these form two levels: an explicit pointwise bound and a direct check on the computed map.

**Proposition E.1** (Pointwise bound under log utility and Cobb–Douglas).

Consider a heterogeneous-household economy with log utility, Cobb–Douglas production (capital share  $\alpha \in (0, 1)$ , depreciation  $\delta \in (0, 1)$ ), and bounded supports for  $(z, a)$ . Let  $R_A^{cs} := 1 + \alpha A(K_A^{cs})^{\alpha-1} - \delta$  and  $r_A^{cs} := R_A^{cs} - 1 = \alpha A(K_A^{cs})^{\alpha-1} - \delta$ . Under Assumption 4 with  $\bar{m}_{\text{path}} = 1$ , the left-hand side of (22) at  $K = K_A^{cs}$  is

$$\text{LHS}(K_A^{cs}) = (1 - \alpha)(r_A^{cs} + \delta) \bar{Z}/L_A + (1 - \alpha)(r_A^{cs} + \delta) \bar{a}/K_A^{cs} - r_A^{cs}, \quad (24)$$

so (22) holds pointwise at  $K_A^{cs}$  whenever (24) is below 1, extending to a neighborhood by continuity; for  $\bar{m} < 1$  it is smaller still.

*Proof.*

At  $K_A^{cs}$ , using  $\alpha A(K_A^{cs})^{\alpha-1} = r_A^{cs} + \delta$ :  $w'_A(K_A^{cs}) = (1 - \alpha)(r_A^{cs} + \delta)/L_A$ ,  $|r'_A(K_A^{cs})| = (1 - \alpha)(r_A^{cs} + \delta)/K_A^{cs}$ ,  $(C_A^{Knull})'(K_A^{cs}) = r_A^{cs}$ . Substituting into (22) with  $\bar{m} = 1$  gives (24); continuity extends the inequality to a neighborhood. ■

In the representative-agent case the Euler condition pins  $R_A^{cs} = 1/\beta$ ; heterogeneity places market-clearing  $R_A^{cs}$  below this. Under stochastic regime-switch expectations  $R_A^{cs}$  instead solves  $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma R_{A'}(K_A^{cs})]$  at  $\Phi_A^{cs}$  (with  $R_{A'}(K_A^{cs}) := 1 + \alpha A'(K_A^{cs})^{\alpha-1} - \delta$ ), generally  $\neq 1/\beta$ ; formula (24) applies in either case.

Proposition E.1 is pointwise plus continuity, not global:  $\sup_{K \in \mathcal{K}^{seg}} \text{LHS}(K) < 1$  still needs numerical verification since  $\bar{Z}, \bar{a}$  depend on the continuation.

**An illustration: feasibility alone does not deliver Assumption 2** Assumption 2 restricts the equilibrium policy, not the feasible set. The budget constraint by itself permits segments that cross the conditional  $K$ -nullcline, so the assumption has content that no resource condition supplies. For non-concave Hill-form production

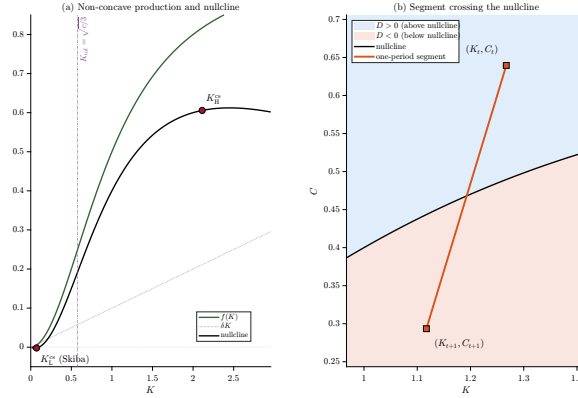
$$f(K) = \frac{A K^2}{c + K^2}, \quad (25)$$

$C^\infty$  with  $f(0) = f'(0) = 0$  and inflection at  $K = \sqrt{c/3}$ , set  $A = c = 1$ ,  $\beta = 0.96$ ,  $\delta = 0.10$ ; the Euler condition  $f'(K) = 1/\beta - 1 + \delta$  has a low unstable root  $K_L^{cs} \approx 0.07$  (convex region) and a Ramsey root  $K_H^{cs} \approx 2.11$  (concave region). Figure E.1 shows the non-monotone nullcline, both steady states, and a budget-feasible segment ( $K_{t+1} = f(K_t) + (1 - \delta)K_t - C_t$ ) that crosses it ( $D_t > 0$ ,  $D_{t+1} < 0$ , so  $D_t D_{t+1} < 0$ ).

This is a geometric possibility, not an equilibrium trajectory: whether the non-concave Ramsey policy selects such transitions requires solving under the non-convex value function. The crossing is also not a consequence of non-concavity. Placing  $C_t$  a fixed gap above the nullcline forces  $K_{t+1} = K_t - \text{gap}$  for *any* production function, since  $f(K_t)$  cancels from the resource constraint, so the same feasible crossing can be built under strictly concave Cobb–Douglas technology. The figure establishes the weaker point that Assumption 2 does not follow from feasibility, and the linearized saddle policy produces no crossings here. What non-concavity supplies is the multiplicity of conditional steady states in panel (a),

which is where monotone convergence is at risk. In standard RBC and [Krusell and Smith \(1998\)](#) economies, concave Cobb–Douglas technology and bounded MPCs make (22) hold with slack.

Figure E.1: Non-concave production: feasibility permits a segment crossing



Notes: Panel (a):  $f(K) = AK^2/(c+K^2)$ ,  $A = c = 1$ ,  $\delta = 0.10$ ; dash-dotted line marks the inflection  $K = \sqrt{c/3}$ ; steady states  $K_L^{cs}$  (unstable) and  $K_H^{cs}$  (Ramsey) on the nullcline  $C = f(K) - \delta K$ . Panel (b): a budget-feasible segment  $(K_t, C_t) \rightarrow (K_{t+1}, C_{t+1})$  crossing the nullcline (blue:  $D > 0$ ,  $\Delta K < 0$ ; red:  $D < 0$ ,  $\Delta K > 0$ ), so  $D_t D_{t+1} < 0$ ; only budget feasibility is illustrated.

## Proof of Theorem E.1 (within-spell $K$ -sufficiency)

*Proof.*

*Part (a): local result.* Near the conditional steady state the distributional dynamics collapse to a one-dimensional linear recursion in the capital gap. Proposition 10 places the orbit tail in a  $C^1$  invariant curve  $\mathcal{W}_A$  tangent to the stable direction; with the aggregate-capital projection (17) of Appendix D, (a)  $\mathcal{W}_A$  is locally homeomorphically parametrized by  $K$  near  $\Phi_A^{cs}$ , and (b)  $k_t := K_t - K_A^{cs}$  obeys  $k_{t+1} = \lambda_A^s k_t + o(|k_t|)$  with  $\lambda_A^s \in (0, 1)$ . Fix  $\varepsilon := \frac{1}{2} \min\{\lambda_A^s, 1 - \lambda_A^s\} > 0$  and a neighborhood  $\mathcal{V}_A$  of  $\Phi_A^{cs}$  on which the parametrization applies and  $|o(|k|)| \leq \varepsilon|k|$ . For  $\Phi_0' \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$  with  $k_t \neq 0$ ,  $(\lambda_A^s - \varepsilon)k_t \leq k_{t+1} \leq (\lambda_A^s + \varepsilon)k_t$  (reversed for  $k_t < 0$ ). Since  $0 < \lambda_A^s - \varepsilon$  and  $\lambda_A^s + \varepsilon < 1$ ,  $k_{t+1}$  shares the sign of  $k_t$  (so the path never crosses  $K_A^{cs}$  and stays in  $\mathcal{V}_A$ ) and  $|k_{t+1}| < |k_t|$ , giving strict monotone geometric convergence to  $K_A^{cs}$ , proving (i). Meanwhile  $\Delta K_t = 0$  (i.e.  $k_{t+1} = k_t$ , so  $\lambda_A^s k_t = k_t + o(|k_t|)$  with  $\lambda_A^s \in (0, 1)$ ) forces  $k_t = 0$ , proving (ii). For (iii), the local homeomorphism makes  $K_t \rightarrow K_A^{cs}$  imply  $\Phi_t \rightarrow \Phi_A^{cs}$ , and main-text Theorem 1 with  $e(\Phi) = K(\Phi)$  yields injectivity of  $K$  on the orbit piece, hence exact conditional sufficiency.

*Part (b): explicit region under Assumptions 3–4.* Under log utility and Cobb–Douglas the benchmark derivatives above identify the functions:  $B_A(K) = \bar{m}(w'_A(K)\bar{Z} + |r'_A(K)|\bar{a})$

inside the sup of (22);  $N_A(K) = (C_A^{Kcnull})'(K)$  inside the inf, making  $B_A - N_A$  its left side; and  $M_A(K) = |(C_A^{Kcnull})''(K)|$  in Lemma E.2. The cap  $\bar{K} \leq (A/\delta)^{1/(1-\alpha)}$  ensures  $C_A^{Kcnull} \geq 0$  on  $[\underline{K}, \bar{K}]$ ;  $K$ -admissibility on  $(\mathcal{K}, \mathcal{V}_A)$  keeps the  $K$ -path and all connecting segments in  $[\underline{K}, \bar{K}]$  (condition (i)).

*Step 1 (endpoint no-side-switching with margin).* The mean-value points  $\xi_t$  (Assumption 4) and  $\xi_t^N$  (Lemma E.1) lie in  $\mathcal{K}$ ; since  $B_A, N_A$  are decreasing,  $B_t - N_t = B_A(\xi_t) - N_A(\xi_t^N) \leq B_A(\underline{K}) - N_A(\bar{K}) \leq 1 - \kappa$ . Lemma E.1 then gives  $D_t D_{t+1} > 0$  for  $K_{t+1} \neq K_t$  and  $|D_{t+1}| \geq (1 + N_t - B_t)|D_t| \geq \kappa|D_t|$ .

*Step 2 (segment no-crossing).* Since  $M_A$  is decreasing, curvature is bounded by  $M_A(\underline{K})$ . Feasibility ( $0 \leq C_t \leq f(K_t; A) + (1-\delta)K_t$ ,  $0 \leq C_A^{Kcnull} \leq f$ ) with the increasing  $2f(K; A) + (1-\delta)K$  and the second endpoint inequality gives  $|D_t| = |K_{t+1} - K_t| \leq 2f(\bar{K}; A) + (1-\delta)\bar{K} < 8\kappa/M_A(\underline{K})$ . Then  $\min\{|D_t|, |D_{t+1}|\} > M_A(\underline{K})|D_t|^2/8$ : (a)  $M_A(\underline{K})|D_t|/8 < \kappa \leq 1$  covers  $|D_t|$ ; (b)  $|D_{t+1}| \geq \kappa|D_t| > (M_A(\underline{K})|D_t|/8)|D_t|$ . Steps 1–2 verify Assumption 2, and sign preservation makes  $\{K_t\}$  strictly monotone.

*Step 3 (sufficiency under admissibility (ii)).* A strictly monotone bounded  $\{K_t\}$ , or a flat step at finite  $t$ , gives a  $K$ -stationary limit with capital in  $\mathcal{K}$ , the limiting state itself supplied by the maintained tightness of  $\{\Phi_t\}$  on  $\mathcal{U}$ ; by (ii) it lies in  $\mathcal{V}_A$  and by part (a)(ii) equals  $\Phi_A^{cs}$  (terminal in the flat-step case). Either way  $K_t \rightarrow K_A^{cs}$  strictly monotonically, and main-text Theorem 1 with  $e = K$  delivers exact conditional sufficiency. ■

**Numerical verification under the Krusell and Smith (1998) calibration (Section 4.1 of the main text)** With  $\alpha = 0.36$ ,  $\delta = 0.025$ , effective productivity  $A \in \{0.944, 0.990\}$  at aggregate labor  $L_A \in \{0.929, 0.970\}$ ,  $\bar{Z} \approx 1$ ,  $\bar{a} \approx \sup_t K_t \approx 38.5$ ,  $\bar{m} \leq 1$ ,  $[\underline{K}, \bar{K}] = [34.6, 38.5]$  (ergodic range):  $B_A(\underline{K}) \approx 0.05$ ,  $N_A(\bar{K}) \approx 0.01$ , so  $B_A(\underline{K}) - N_A(\bar{K}) \approx 0.041 < 0.05 = 1 - \kappa$  with  $\kappa = 0.95$ . For curvature,  $M_A(\underline{K}) \approx 6.8 \times 10^{-4}$  gives  $8\kappa/M_A(\underline{K}) \approx 1.1 \times 10^4$ , while  $2f(\bar{K}; A) + (1-\delta)\bar{K} \approx 2(3.7) + 37.5 \approx 45$ : a margin over two orders of magnitude. Here  $\bar{Z} \leq z_{\max}$  is a priori from compact support;  $\bar{a}$  and the ergodic range are read off the computed equilibrium, so the check validates the continuation rather than replacing computation. Replacing  $\bar{a}$  by the feasibility cap  $(A/\delta)^{1/(1-\alpha)} \approx 314$  still gives  $B_A(\underline{K}) - N_A(\bar{K}) \approx 0.23 < 1$ .

**Why the residual condition cannot be closed from primitives** Admissibility condition (ii) requires no  $K$ -stationary state with capital in  $\mathcal{K}$  to lie outside  $\mathcal{V}_A$ . Closing it from primitives means characterizing the basin of attraction of the conditional steady state, ruling out moving plateaus where aggregate net saving vanishes while  $\Phi$  keeps evolving. The MPC

bounds control aggregate consumption relative to capital but is silent when both vanish, so a primitive treatment would need either strict monotonicity of aggregate saving in a distributional ordering or a global Lyapunov function for  $\Gamma_{\text{endo}}(\cdot, A)$ , both beyond scope. The condition is checkable along any computed continuation and fails to bind in every calibration examined.

## F Continuous dependence on beliefs

This appendix establishes continuity of the finite-horizon frozen-regime continuation in the belief parameter  $\mu$ , where  $\Pi(\mu) = (1 - \mu)I + \mu\Pi$ , so that  $\mu = 0$  is perfect foresight and  $\mu = 1$  rational expectations. It does not require the ergodic distribution of the full stochastic process.

**Lemma F.1** (Continuity of the frozen-regime continuation in beliefs).

*Fix an initial distribution  $\Phi_0$ , a regime  $A$ , and a finite horizon  $T$ . Suppose the asset domain is compact, wages are bounded away from zero, and the period return is bounded and continuous on the feasible set, with  $u$  strictly concave. Then the frozen-regime iterates  $\{\Phi_t(\mu)\}_{t=0}^T$  and the induced aggregates  $\{(K_t(\mu), C_t(\mu))\}_{t=0}^T$  are continuous in  $\mu$  on  $[0, 1]$ .*

*Proof.*

*Step 1 (Household policies).* With  $\Pi(\mu) = (1 - \mu)I + \mu\Pi$  entering the continuation expectation linearly, the Bellman operator  $T_\mu$  for  $v(a, z, A; \mu)$  on the compact  $[a, a_{\max}]$  is a sup-norm contraction for each  $\mu$  and is continuous in  $\mu$ . The hypotheses of [Stokey et al. \(1989, Theorem 9.6\)](#) hold: the feasible correspondence is nonempty, compact-valued and continuous, and the period return is bounded by hypothesis. By the parametric contraction mapping theorem its fixed point  $v(\cdot; \mu)$  depends continuously on  $\mu$ , and strict concavity makes the policy single-valued and continuous ([Stokey et al., 1989, Theorem 9.8](#)), so by the maximum theorem  $c(\cdot, A; \mu)$  and  $a'(\cdot, A; \mu)$  do too.

*Step 2 (Frozen-regime operator).*  $\Gamma_{\text{endo}}(\cdot, A; \mu)$  composes the continuous price map  $\Phi \mapsto (r(\Phi, A), w(\Phi, A))$ , the jointly continuous Step 1 policy, and the push-forward to next-period distribution, so it is jointly continuous in  $(\Phi, \mu)$  (distributions under weak convergence,  $\mu$  under Euclidean distance).

*Step 3 (Finite iterates).* For fixed  $\Phi_0, T$ ,  $\Gamma_{\text{endo}}^t(\Phi_0, A; \mu)$  composes  $t$  Step 2 maps, so  $\{\Phi_t(\mu)\}_{t=0}^T$  and the aggregates  $\{(K_t(\mu), C_t(\mu))\}_{t=0}^T$  are continuous in  $\mu$  pointwise in  $t$  for every finite  $T$ .

■



## G Order-preservation and within-spell monotonicity

Within a frozen regime the aggregate law of motion reduces to a scalar map  $g_A : K_t \mapsto K_{t+1}$ , the aggregate savings response with the regime held at  $A$ . Whether aggregate capital indexes the equilibrium state within the spell is governed by whether this map preserves order.

**Proposition G.1** (Within-spell monotonicity under order-preservation).

*Fix  $(\Phi_0, A)$ . If  $g_A$  is continuous and strictly increasing on the relevant interval, with a unique fixed point  $K_A^{cs}$  at which  $g'_A(K_A^{cs}) \in (0, 1)$ , then aggregate capital converges strictly monotonically to  $K_A^{cs}$  along the frozen-regime continuation from every  $K_0$  in the interval.*

*Proof.* Strict monotonicity of  $g_A$  with a unique fixed point makes  $g_A(K) - K$  single-signed on each side of  $K_A^{cs}$ ; with  $g'_A(K_A^{cs}) < 1$ ,  $g_A$  lies above the 45° line for  $K < K_A^{cs}$  and below it for  $K > K_A^{cs}$ . Hence  $K_0 < K_A^{cs}$  gives  $K_0 < g_A(K_0) < K_A^{cs}$ , an increasing sequence bounded above by  $K_A^{cs}$ ; the case  $K_0 > K_A^{cs}$  is symmetric. Order-preservation keeps the iterates on one side of  $K_A^{cs}$ , so the path cannot overshoot and converges monotonically. ■

Order-preservation of  $g_A$  is a direct, checkable property of the computed policy, not an inference from a belief homotopy. Proposition G.1 applies where the frozen-regime law is scalar: the representative-agent RBC of Appendix J. In the Krusell and Smith (1998) economy  $K_{t+1}$  is a functional of  $\Phi_t$ , so no scalar  $g_A$  is available in advance and the check is made pathwise instead: all 84 realized frozen-regime spells are strictly monotone in  $K$ , which is the hypothesis main-text Proposition 8 uses. The rank correlations of  $K_{t+1}$  on  $K_t$ , 0.9999 and 1.0000, describe the realized scatter and are consistent with order-preservation without establishing it. It is the structural driver of within-spell monotonicity, plausibly belief-independent under concave savings and monotone price feedback, and cheaper to certify than a global uniqueness argument. With the curvature-margin check (23) of Appendix E, order-preservation delivers the segment condition of Assumption 2, and the argument extends to any continuous aggregate  $e$  with unique stationarity, covering the CRRA–separable-labor extension.

Perfect-foresight monotonicity does not by itself transport to rational expectations. Writing  $\Pi(\mu) = (1 - \mu)I + \mu\Pi$  for the belief homotopy ( $\mu = 0$  perfect foresight,  $\mu = 1$  rational expectations), a path strictly monotone at  $\mu = 0$  can develop a discrete-time overshoot as  $\mu$  increases, because path monotonicity is not preserved under continuous perturbation once the map loses order-preservation.<sup>3</sup> The object to verify is therefore order-preservation

<sup>3</sup>For a scalar illustration,  $g_\mu(x) = \frac{1}{2}x - \frac{3}{2}\mu x^3/(1+x^2)$  has a unique stable fixed point at 0 with  $g'_\mu(0) = \frac{1}{2}$



of the map itself, for which the perfect-foresight map is the cheapest instance to inspect.

## H Calibrating the asymmetric adjustment cost

The asymmetric adjustment cost has two curvatures,  $\mu_+$  (expansion) and  $\mu_-$  (contraction), reparameterized as a level  $\bar{\mu} = (\mu_+ + \mu_-)/2$  and an asymmetry  $r = \mu_+/\mu_-$ ; the other annual parameters are  $\beta = 0.96$ ,  $\delta = 0.10$ ,  $\alpha = 0.33$ , and  $A \in \{0.98, 1.02\}$  with transition diagonal 0.90. Moments use annual log first-differences of real chained-dollar FRED series: private nonresidential fixed investment (PNFICA, 1967–2019), gross private domestic investment (GPDICA, 1947–2019), and GDP, all pre-COVID.

**Asymmetry  $r$**  The friction makes contractions sharper than expansions, so the target is the negative skewness of investment growth. The third-moment coefficient is unreliable in a 52-observation sample dominated by 2009 (−1.03 full-sample, −0.54 dropping 2009, −0.32 for GPDICA), so I target the outlier-robust downside/upside semideviation ratio  $\sigma_-/\sigma_+$  (RMS below- over above-mean deviations of  $\Delta \log I$ ): 1.32 (PNFI), 1.11 (GPDICA), both above one. The model ratio rises monotonically with  $r$  but saturates near 1.30, so the band bounds  $r$  from below and not from above. Symmetry and reversed asymmetry are excluded, since  $r \leq 1$  delivers 1.01 or less; the lower edge 1.11 is crossed at  $r = 2.31$ ; and the PNFI value 1.32 is not attained by any finite  $r$ . The moment therefore identifies a half-line,  $r \gtrsim 2.3$ , rather than a point. I set  $r = 3$ , the smallest round value in that set, hence  $(\mu_-, \mu_+) = (1.53, 4.57)$ , with  $\mathcal{R} \in [1.16, 1.34]$  across  $r \in [2, 4]$  (1.265 at  $r = 3$ ). This is a conservative conditional choice, not a structural estimate, and the level of  $r$  within the identified set is not disciplined by this moment.

**Level  $\bar{\mu}$**  Relative investment volatility cannot identify the level. Even at negligible friction the  $\pm 2\%$  TFP process reaches only  $\text{sd}(I)/\text{sd}(Y) \approx 3.12$ , which matches PNFI (3.21, s.e. 0.38) but falls far short of GPDICA (4.56). The moment is therefore uninformative about the level of  $\bar{\mu}$ : the frictionless limit already sits inside one target’s standard error, so friction cannot be identified by driving the model down to it. I anchor  $\bar{\mu}$  on the investment– $q$  elasticity:  $q = 1 + \bar{\mu}(K' - K)/K$  gives a net-investment-rate response to  $q$  of  $1/\bar{\mu}$ , and at  $\bar{\mu} = 3.05$ ,  $r = 3$  the curvatures imply  $1/\bar{\mu} \in [0.22, 0.65]$ , within measurement-corrected estimates. The level barely affects  $\mathcal{R}$ .

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and  $\Delta x = 0$  only there. The path from  $x_0 = 1$  is  $1, \frac{1}{2}, \frac{1}{4}, \dots$  at  $\mu = 0$  but  $1, -\frac{1}{4}, -0.10, \dots$  at  $\mu = 1$ : strictly monotone in the first case, overshooting the steady state in the second.

Table H.1: Asymmetric adjustment costs: robust-asymmetry calibration and same-capital slopes

| Case                               | $\mu_-$ | $\mu_+$ | $\sigma(I^{cyc})/\sigma(Y^{cyc})$ | $\sigma_-/\sigma_+(\Delta \log I)$ | skew( $\Delta \log I$ ) | med $\mathcal{R}(K)$ | $C/Y$   |
|------------------------------------|---------|---------|-----------------------------------|------------------------------------|-------------------------|----------------------|---------|
| Data                               | —       | —       | 3.21                              | 1.318                              | -1.03                   | —                    | —       |
| Semideviation-targeted ( $r = 3$ ) | 1.525   | 4.575   | 1.85                              | 1.140                              | -0.63                   | 1.26                 | 0.0032% |
| Same-mean symmetric                | 3.050   | 3.050   | 1.79                              | 1.014                              | -0.07                   | 0.98                 | 0.0033% |
| Same-mean sign reversal            | 4.575   | 1.525   | 1.81                              | 0.899                              | 0.51                    | 0.75                 | 0.0032% |

*Notes:* The asymmetry ratio  $r = 3$  is disciplined by the outlier-robust downside/upside semideviation ratio of  $\Delta \log(\text{PNFICA})$  over 1967–2019; the moment skewness is reported for reference only, as it is fragile to the 2009 outlier.  $I^{cyc}$  and  $Y^{cyc}$  are annual HP-filtered log investment and output with  $\lambda = 6.25$ ; their relative volatility is untargeted and cannot identify the level.  $\mathcal{R}(K) = C'_B(K)/C'_G(K)$  is summarized by its median on the common interior capital support, and its ordering across the three rows is the comparative static reported in Section 3 of the main text.  $C/Y$  is the mean resource cost.

## I Diagnostic validation under lumpy investment

This appendix runs the boundary test of Section 4.1, where nullcline invariance fails and  $K$ -sufficiency is no longer a theorem: does the monotonicity diagnostic still track scalar-state forecast accuracy? Three economies are solved with the same global method: [Khan and Thomas \(2008\)](#) in general equilibrium, the same model with prices frozen at stationary-equilibrium values (where lumpiness is known to matter), and the [Krusell and Smith \(1998\)](#) benchmark. For each frozen-regime spell of length  $\geq 3$  I compute two statistics. The first is the monotonicity diagnostic  $|\sum_t \Delta K_t|/\sum_t |\Delta K_t|$ , which is one iff capital moves monotonically within the spell (Proposition 8; spells with a  $\Delta K$  sign change are *flagged*). The second is the scalar-state forecast error from a regime-specific rule  $\log K' = b_0(A) + b_1(A) \log K$  iterated dynamically within the spell ([Den Haan, 2010](#)), recorded as the maximum log-deviation from the realized path.

**Results** Where scalar-state accuracy is high, the diagnostic agrees without exception: all 215 spells of the [Khan and Thomas \(2008\)](#) general-equilibrium economy and all 84 of the [Krusell and Smith \(1998\)](#) benchmark are exactly monotone in  $K$ , with dynamic errors averaging  $4.4 \times 10^{-4}$  and  $5.3 \times 10^{-4}$  in logs. Price movements thus smooth aggregate investment enough to keep the within-spell path monotone, rationalizing approximate aggregation under lumpy investment.

Removing price feedback breaks accuracy sharply: mean dynamic errors rise about twenty-sevenfold, to  $1.2 \times 10^{-2}$ , while the one-step  $R^2$  stays at 0.998, so the one-step fit is uninformative about dynamic accuracy ([Den Haan, 2010](#)). Monotonicity, by contrast, is a blunter signal: of the 127 frozen-price spells only one is flagged, and the median spell is still

exactly monotone. Controlling for spell length, flagged spells carry a log-error premium of 1.56, and the correlation between 1-monotonicity and the log dynamic error is 0.24. The flag is thus necessary rather than sufficient here: a monotonicity failure signals trouble, but monotonicity alone does not certify accuracy once nullcline invariance fails and the distribution carries independent forecasting power ([Bachmann et al., 2013](#)).

## J Assumptions 1 and 2 in the representative-agent RBC

In the representative-agent RBC, Assumptions 1 and 2 follow from the scalar concave structure. The quantitative inputs are the Step 4 curvature-margin condition and, for stochastic beliefs, the Step 2c single-crossing condition on the conditional Euler residual. The endogenous state is scalar capital  $K$ , with frozen- $A$  law of motion

$$g_A(K) := \Gamma_{\text{endo}}(K, A), \quad (26)$$

the equilibrium savings policy of the stochastic RCE with the regime held at  $A$ . Production is Cobb–Douglas  $f(K; A) = AK^\alpha$ ,  $\alpha \in (0, 1)$ , depreciation  $\delta \in (0, 1)$ , utility strictly concave and Inada, and the regime a finite irreducible Markov chain.

**Proposition J.1** (Geometric regularity holds in the canonical RBC).

*Under the maintained Cobb–Douglas/strictly-concave-utility specification, together with the curvature-margin condition of Step 4 and, for stochastic beliefs, the verified single-crossing condition of Step 2c, Assumptions 1 and 2 hold for the representative-agent RBC. For each  $A$ :*

- (i) *the frozen- $A$  continuation converges to a unique conditional steady state  $K_A^{cs}$ , and aggregate capital is strictly  $K$ -monotone along it;*
- (ii) *the continuation makes no segment crossing of the conditional  $K$ -nullcline.*

*The curvature-margin condition is used only for (ii); the single-crossing condition only for the stochastic-beliefs case of (i). The deterministic-beliefs conclusions follow from strict concavity alone.*

*Proof.*

*Step 0 (Policy selection).* The stochastic RCE solves a strictly concave dynamic program, so the optimal policy is single-valued and continuous in  $(K, A)$  ([Stokey et al., 1989](#), Theorem 9.8); interior  $C^1$  smoothness follows from [Santos \(1991\)](#) and is used only locally at

$K_A^{cs}$ . Restricting this policy to fixed  $A$ ,  $g_A$  is a well-defined continuous self-map of the capital domain. It is increasing because  $u(f(K; A) + (1 - \delta)K - K')$  has increasing differences in  $(K, K')$ , since  $u$  is concave and  $f'(K; A) + 1 - \delta > 0$ . This gives Assumption 1(ii).<sup>4</sup>

*Step 1 (Feasibility rules out explosion).* Any feasible continuation satisfies  $K_{t+1} \leq f(K_t; A) + (1 - \delta)K_t$ , and for  $K_t > \bar{K}_A := (A/\delta)^{1/(1-\alpha)}$  diminishing returns give  $f(K_t; A) < \delta K_t$ , hence  $K_{t+1} < K_t$ . So  $K_t \leq \max\{K_0, \bar{K}_A\}$  on every branch, a feasibility property needing no transversality condition.

*Step 2 (Unique conditional steady state).* (2a) Fixing  $\underline{K} \in (0, \bar{K}_A)$  with  $g_A(\underline{K}) \geq \underline{K}$  (exists by Inada), the increasing  $g_A$  maps  $[\underline{K}, \bar{K}_A]$  into itself (Step 1), so a fixed point  $K_A^{cs} = g_A(K_A^{cs})$  exists (Brouwer). (2b) As the policy of a strictly concave one-sector problem,  $g_A$  induces monotone convergence to a unique steady state: the turnpike property (Stokey et al., 1989, Ch. 6), i.e.  $g_A(K) - K$  single-crosses zero. At  $\mu = 0$  this is the deterministic Ramsey result. (2c) At  $\mu = 1$  existence and monotonicity are inherited from concavity as in (2a)–(2b). Global uniqueness and monotone convergence need one further input: the conditional Euler residual at  $\mu = 1$  crosses zero exactly once on the computed range, so  $g_A(K; 1) - K$  changes sign once. This single-crossing is not delivered by concavity; it is verified in the computed equilibrium (Appendix A: a single crossing from above of each conditional Euler residual (1) at  $\mu = 1$ ). Uniqueness rests on the direct  $\mu = 1$  check; Appendix F supplies only the continuity of the finite-horizon continuation in  $\mu$ .

This gives Assumption 1(i), and unique  $K$ -stationarity follows because next period's  $K$  depends on  $K$  alone, so  $\Delta K_t = 0$  only at the fixed point.

*Step 3 (Strict monotonicity; endpoint no-side-switching).* With  $g_A$  increasing, slope  $\lambda_A^s \in (0, 1)$  at  $K_A^{cs}$ , and Step 2 single-crossing, the iterates  $\{K_t\}$  are monotone toward  $K_A^{cs}$  (from any  $K_0$  under deterministic beliefs; from the verified range under stochastic beliefs). The identity  $\Delta K_t = -D_t$ ,  $D_t := C_t - C_A^{Knull}(K_t)$ , then keeps  $D_t$  of constant nonzero sign before convergence:  $D_t D_{t+1} > 0$ .

*Step 4 (Segment no-crossing).* By the curvature-margin lemma (Lemma E.2), endpoint no-side-switching upgrades to segment no-crossing whenever  $\min\{|D_t|, |D_{t+1}|\} > M(K_{t+1} -$

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<sup>4</sup>Strict concavity, not transversality, rules out the frozen-branch overaccumulation path; transversality only selects the stable recursive policy. The necessary condition is the pooled (expectation) condition (Kamihigashi, 2003, 2005), refining to a regime-conditional one under a finite chain; neither disciplines the frozen- $A$  branch, a probability-zero event. The all- $A$  overaccumulation trajectory ( $K_t \rightarrow \bar{K}$ ,  $C_t \rightarrow 0$ ,  $u'(C_t) \rightarrow \infty$ ) is capital-bounded and violates the branchwise condition yet is hidden in expectation by its weight  $\pi_{AA}^T$ ; strict concavity excludes it as suboptimal at every positive-probability finite-horizon node.

$K_t)^2/8$ , with  $M = \sup \alpha(1 - \alpha)AK^{\alpha-2} < \infty$  on the compact saddle. The RBC identity  $|K_{t+1} - K_t| = |D_t|$  reduces this to  $|D_{t+1}| > M|D_t|^2/8$ . The curvature-margin condition is therefore that there exist  $\kappa > 0$  and  $\bar{D}$  with  $|D_{t+1}| \geq \kappa|D_t|$  at every pre-limit transition and  $\sup_t |D_t| = \bar{D} < 8\kappa/M$ . Step 1 bounds  $\bar{D}$  by feasibility, though whether that bound falls below  $8\kappa/M$  is a quantitative condition checked in the calibration, not a consequence of feasibility. The first inequality holds whenever  $g_A$  is differentiable with  $\inf g'_A > 0$  on the compact saddle, by the mean value theorem, which Step 0 monotonicity alone does not supply; it is the scalar counterpart of the  $|D_{t+1}| \geq \kappa|D_t|$  step of Theorem E.1(b). Steps 3–4 give Assumption 2. ■

The key input is strict concavity: it selects the policy (Step 0), bounds capital (Step 1), and yields the unique conditional steady state and monotone convergence (Steps 2–3); the stochastic case adds the verified single-crossing (Step 2c) and the feasibility-bounded curvature closing Step 4. The heterogeneous-agent economy lacks this. There  $\Gamma_{\text{endo}}$  acts on the distribution, and next period's  $K$  depends on it rather than on  $K$  alone. The stable-eigenvalue property then belongs to an infinite-dimensional operator, and primitives do not determine it. The non-concave illustration of Appendix E shows that feasibility alone does not deliver Assumption 2. What concavity supplies is the unique conditional steady state and the monotone policy, and dropping it puts both at risk.

## K The investment-externality example

This appendix adds investment-generated knowledge  $H$  to Section 2 RBC, so  $\tilde{A} = AH^\eta$ . Only  $A$ , not  $H$ , is frozen. The replication materials contain the solution and figure code; [De Long and Summers \(1991\)](#) and [Irwin and Klenow \(1994\)](#) motivate the equipment-return and learning-spillover channels.

**Corrected planner system and the flow incentive** Because investment jointly produces  $K'$  and  $H'$ , the standard substitution  $p_{t+1} = \lambda_{t+1}(1 - \delta + \alpha Y_{t+1}/K_{t+1})$  no longer holds, and the planner's two co-states  $p = V_K$  and  $q = V_H$  must be carried explicitly (main text, Section 4.2). Combining the Euler and capital-envelope conditions gives the shadow value of installed capital,  $p_t = \lambda_t(\alpha Y_t/K_t + 1 - \delta) - (1 - \delta)g'(I_t)\beta\mathbb{E}_t[q_{t+1}]$ , *below* its consumption value: installed capital lets the planner cut current gross investment while holding  $K_{t+1}$  fixed, forgoing knowledge production, so the knowledge incentive nets to a flow. In a deterministic

steady state the system reduces to

$$1 = \beta \left[ \frac{\alpha Y/K}{1 - \beta(1 - \delta)} + \frac{g'(I) \eta Y/H}{1 - \beta(1 - \delta_H)} \right], \quad (27)$$

and the steady-state-matching credit is the flow incentive itself,  $\bar{\tau} = g'(I)\beta q/\lambda$ , that is,  $g'(I)\beta(\eta Y/H)/[1 - \beta(1 - \delta_H)]$ , with no further capitalization. At the calibration this yields  $K^{**} = 39.49$  (+3.9%),  $H^{**} = 1.027$  (+2.7%), and  $\bar{\tau} = 0.0242$ . Eliminating  $Y/K$  between this expression and the planner's unit-investment condition (27) gives the closed form  $\bar{\tau} = bm/(a + bm)$ , with  $m = \eta\phi$ ,  $a = \alpha/[1 - \beta(1 - \delta)] = 10.3597$  and  $b = \delta_H/\{\delta[1 - \beta(1 - \delta_H)]\} = 36.6972$ . The normalizing constant  $\gamma$  cancels through  $\delta_H H = \gamma I^\phi$ , so  $\bar{\tau}$  depends on  $(\eta, \phi)$  only through  $\eta\phi$ , with  $\bar{\tau}/(1 - \bar{\tau})$  proportional to it and  $d \ln \bar{\tau}/d \ln(\eta\phi) = 1 - \bar{\tau}$  identically. Halving and doubling  $\eta$  gives 0.01225 and 0.04725, so proportionality holds to about two percent.

**Conditional steady states by frozen-regime solves** The conditional steady states come from dedicated frozen-regime RTM solves that hold the realized path in one regime forever while expectations still price switches. The stay branch uses the path's own future values, so interpolation enters only through the five-percent switch branch, which cross-regime corridors keep inside sampled support by sampling each regime's policy around the *other* regime's steady state. The two conditional steady states, (37.07, 0.983) and (38.92, 1.017), bracket the ergodic cloud ( $K \in [37.10, 38.91]$ ) asymmetrically, the bad-regime point just outside the lower edge and the good-regime point on the upper edge: the longest good spells essentially attain their steady state, while bad spells stop just short. The planner's conditional steady states, (38.54, 1.010) and (40.43, 1.045), come from the same construction applied to its co-state system and bracket the planner's ergodic range [38.59, 40.43] with the same asymmetry. Anchors placed on opposite sides of every steady state agree to  $10^{-3}$  of capital or better, and a local-linear switch branch moves the points by at most 0.07 percent.

**Saddle counting** Table K.1 reports the counts and eigenvalue magnitudes behind the main-text saddle classification. Finite-difference Jacobians of the frozen switch-priced map at the two conditional steady states place the bad-regime roots cleanly inside the unit circle. In the good regime the slow root sits along a near-neutral direction where differencing an interpolated surface is at its resolution limit: its estimated magnitude spans unity (0.98–1.04), so its local classification is ambiguous. Dedicated frozen paths still converge to

the steady state (decay rates 0.94 and 0.96), establishing attraction without identifying the second local eigenvalue.

Table K.1: Stable and unstable roots: deterministic benchmark and frozen maps

| System (variables)                     | Stable | Unit | Unstable | Eigenvalue magnitudes                 |
|----------------------------------------|--------|------|----------|---------------------------------------|
| Equilibrium, concave $(K, H, C)$       | 2      | 0    | 1        | 0.903, 0.965   1.047                  |
| Planner, concave $(K, H, C, q, p)$     | 2      | 0    | 3        | 0.900, 0.966   1.045, 1.122, $\infty$ |
| Equilibrium, homogeneous $(K, H, C)$   | 1      | 1    | 1        | 0.986   1.000000   1.169              |
| Frozen map at steady state, regime $B$ | 2      | 0    | 0        | 0.90–0.94, 0.98–0.99                  |
| Frozen map at steady state, regime $G$ | 1–2    | 0    | 0–1      | 0.93–0.95, 0.98–1.04                  |

**Validation of the interpolated surface** The surface is interpolated from the ergodic cloud plus thirty-four anchored corridors and validated under both the solver’s kNN interpolant and an independent local-linear estimator. Off-path Euler residuals, computed from an Euler equation the fit never targeted, have interior median  $1.1 \times 10^{-5}$  under both estimators, rising only in a band along the support boundary where interpolation is one-sided. Leaving each corridor out and re-predicting its solved consumption from the rest reproduces it with worst-corridor mean  $1.3 \times 10^{-5}$ . The planner surface is validated the same way, with interior residuals at medians up to  $2.4 \times 10^{-5}$ . On the common support the planner consumes less at all 521 evaluated states, by 1.3–2.9 percent (mean 1.8), and the gap correlates with the marginal knowledge return  $g'(I)$  at 0.980 across the good-regime grid. This is the correction varying, not extra volatility: on a common shock path the planner’s ergodic consumption is 0.6 percent higher than the equilibrium’s at an essentially equal log standard deviation.

**The homogeneous cone and hysteresis** In the homogeneous benchmark ( $\phi = 1, \eta = 1 - \alpha$ ) the conditional saddle surface is a cone. Degree-one homogeneity is verified by a level-space shooting solve that nowhere imposes the scaling symmetry, to mean error  $9.5 \times 10^{-11}$ . The neutral ray carries permanent level effects: paths from ratios  $0.8x^*$  and  $1.2x^*$  at the same initial knowledge converge to the common rest ratio  $x^* = 37.99$ , with knowledge permanently at 1.156 against 0.851. This rest ratio equals the concave-case equilibrium capital  $K^*$ , since both solve  $1 = \beta(1 - \delta + \alpha x^{\alpha-1})$  at  $A = 1$ . The regime-conditional rest ratios are  $x_B^{\text{cs}} = 42.52 > x_G^{\text{cs}} = 35.04$ , bracketing the ergodic cluster, and  $x$  is strictly monotone in all 642 realized spells (main text, Section 4.2).

**State-contingent versus constant credit** The state-contingent credit  $\tau_t$  aligns the surfaces exactly. Fed back through the credited household Euler on the computed planner path, it reproduces the planner’s allocation to a mean residual of  $2.8 \times 10^{-6}$ , a numerical decentralization rather than only the algebraic identity. The constant  $\bar{\tau} = 0.0242$  matches only the deterministic  $A = 1$  planner steady state and has not been shown to make the stochastic surfaces coincide at any state. It nevertheless tracks the planner’s transition within 0.1 percent in capital, knowledge, and consumption over eight hundred quarters from a common anchor, because the state-contingent credit barely moves along that corridor,  $\tau_t \in [0.0237, 0.0247]$ . The distinction should grow when  $g'$  or the knowledge co-state varies strongly along transitions, for example toward the composite-returns boundary  $\alpha + \eta\phi = 1$  (main text, Section 4.2).

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